

Concept Plans and Spatial Organization Diagrams

How the park is organised — the crescent, the radial alleys, and the rooms between them.

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- Phase4 Concept Development Report
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VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	src/plan.py · src/drawings.py
Reproduce	<code>python run_analysis.py</code> · <code>python -m tests.test_pipeline</code>

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/plan.py](#)

[src/drawings.py](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

Design Narrative & Concept

Falaj Al Safa — an arc of shade for a park that is currently too hot to use

Of the 4,402 daylight hours in a Dubai year, only **44.5%** are comfortable to stand in on the open site today. This proposal raises that to **64.6%** — a gain of 20.1 percentage points, or about 884 additional usable hours a year.

Upload slot 01 — Design Narrative & Concept

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The problem, stated as a number

A neighbourhood park nobody can stand in is not a park. Al Safa 2's difficulty is not layout, planting or maintenance — it is thermal. The site is flat and largely exposed, at 25°N, where the summer sun is nearly overhead and the winter sun is low enough to slide underneath most shade.

The analysis behind this submission reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, verified back against those normals to within 0.39 °C. Across the 4,402 daylight hours in that year, the open site is comfortable for 44.5% of them. That is the number the design exists to move.

2. The concept

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in `src/plan.py`.

3. Why an arc — the part that was solved, not styled

A straight canopy presents one orientation. When a sun angle defeats it, it defeats the whole length at once, and the walk has no shade anywhere along it. An arc changes heading continuously, so some segment is always angled well.

The depth of the bow was not chosen for appearance. It was swept against the 8,760-hour solar model at a fixed section, over all 4,402 daylight hours of the year. The criterion was the final column — the hours in which the route offers nowhere at all to stand.

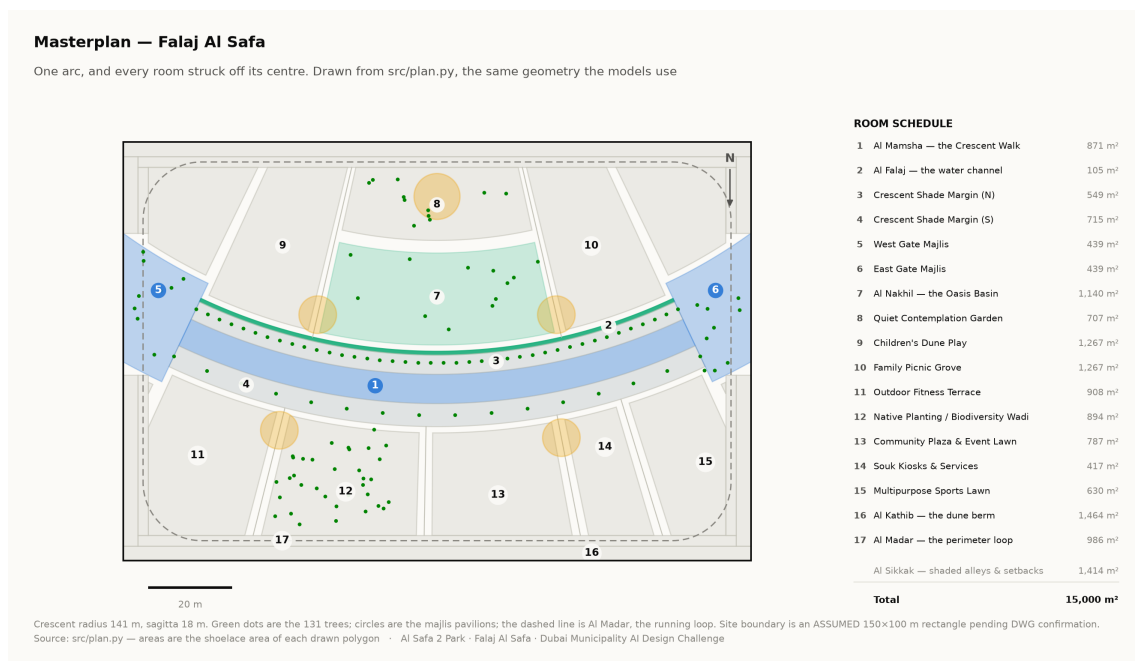
Plan form	Mean cover	Worst month	Hours with no shade anywhere
Straight east–west bar	87.4%	68.7%	330
Sine meander (superseded)	85.0%	73.1%	63
Arc, sagitta 10 m	87.1%	70.2%	116
Arc, sagitta 14 m	86.6%	71.3%	62
Arc, sagitta 18 m	85.9%	72.1%	52
Arc, sagitta 22 m	84.9%	72.3%	61
Closed elliptical loop	79.1%	69.8%	89

Adopted row highlighted. Source: the sweep recorded in `src/config.py`, measured against the same solar model used throughout this submission.

Read the first column carefully. The straight bar has the *highest* mean coverage of anything tested — an east–west canopy is close to the optimum orientation for 25°N, and curving it costs about a point. The crescent is not claimed to shade more ground on average. It is claimed to remove the hours in which the route offers nowhere at all to stand, and it removes six sevenths of them. That is the trade, stated in the direction that is not flattering.

The closed loop loses on every measure, for the same reason the deep bows do: it forces half its length to run north–south, and a canopy over a north–south route can only work when the sun is low in the east or west. The loop survives in the scheme as Al Madar, an *unshaded* running circuit — because that is what a circuit is actually for.

The arc bows convex south. The solar model is indifferent to that sign, so the choice was spatial: bowing south puts the structure between the sun and the hollow it wraps, making the concave side the park's cool pocket instead of a south-facing bowl. Every room people are asked to linger in sits in that hollow.



Masterplan. Every area is the shoelace area of the drawn polygon, which is why the schedule closes on 15,000 m² without anyone reconciling a spreadsheet. Technical drawing — to scale.

4. What the design achieves

Measure	Today	As designed
Comfortable daylight hours	44.5%	64.6%
Peak heat index	56.8 °C	48.7 °C
Mean heat-index reduction under canopy	—	7.13 °C
Crescent Walk shaded (canopy and louvre)	—	87.3%
Site-wide mean shade	—	34.1%
Trees	—	131

Site-wide mean shade is modest at 34.1%, and that is a position, not an oversight. This scheme makes a few places genuinely excellent rather than the whole site marginally better. A park with even shade everywhere and no continuously comfortable route would score higher on that one statistic and be worse to use.

5. Two corrections this submission makes to itself

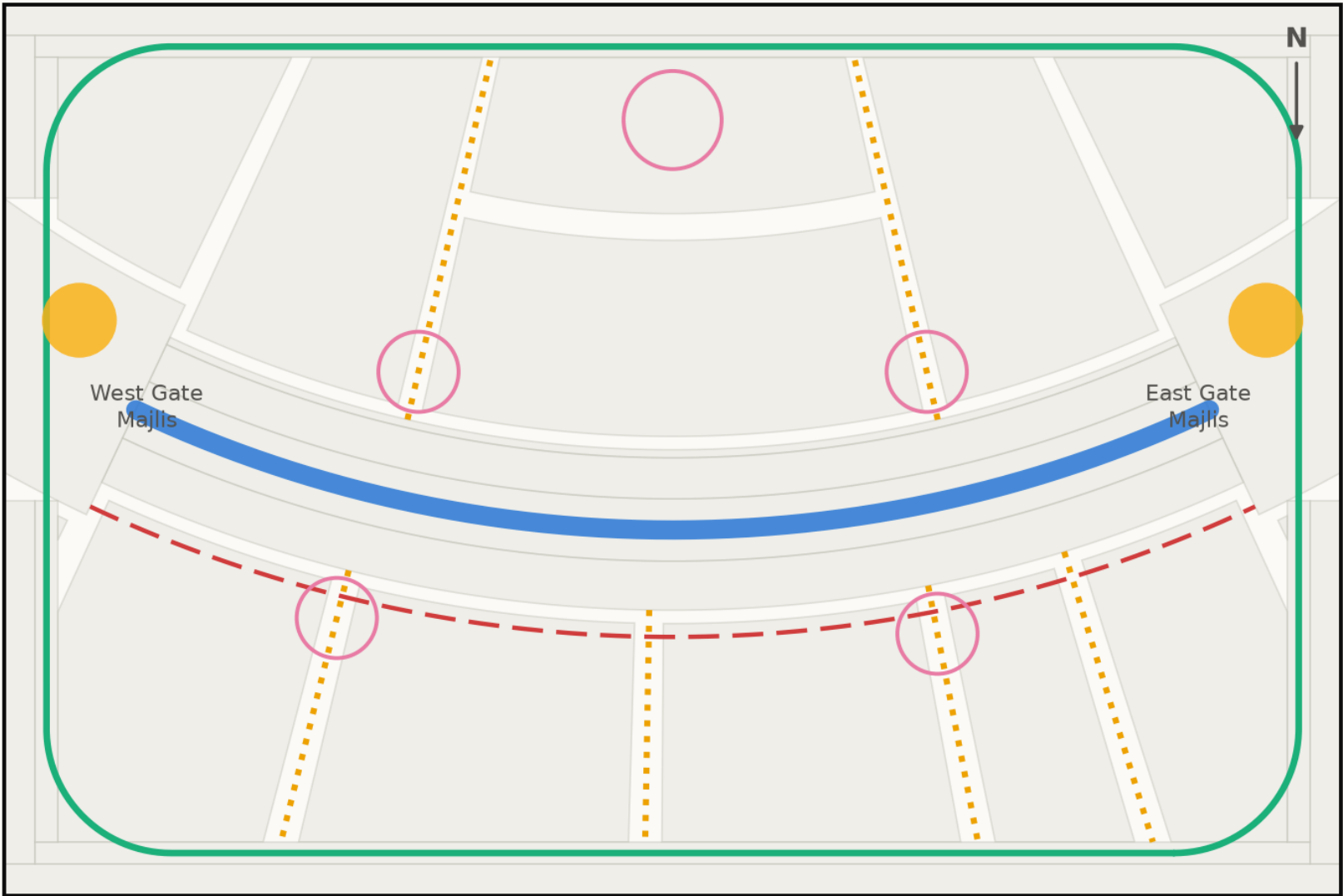
A withdrawn shade claim. An earlier version claimed 99.2% annual shade on a flat 9 m canopy over a 9 m walkway. It does not survive a geometric check — when the sun is low and to the south, the shadow slides clean off the path — and it was withdrawn. The section was then re-solved: a narrower walk, a wider overhang, a lower plane and a deep southern louvre. It now measures 87.3%, and that is a number that can be checked.

Three withdrawn images. Three visuals presented invented data as measurement: a thermal comfort analysis attributed to CFD when no CFD was run; a satellite NDVI analysis whose raster was noise; and a parametric canopy described as solar-optimised when no objective was ever defined. All three were withdrawn on the same principle as the 99.2%: anything that cannot survive being checked should not be in front of a juror.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free
- Al Sikkak — shaded alleys to the rooms
- Al Madar — 438 m running loop
- Service / emergency vehicle access
- Majlis — shaded rest, never more than 33 m from the walk

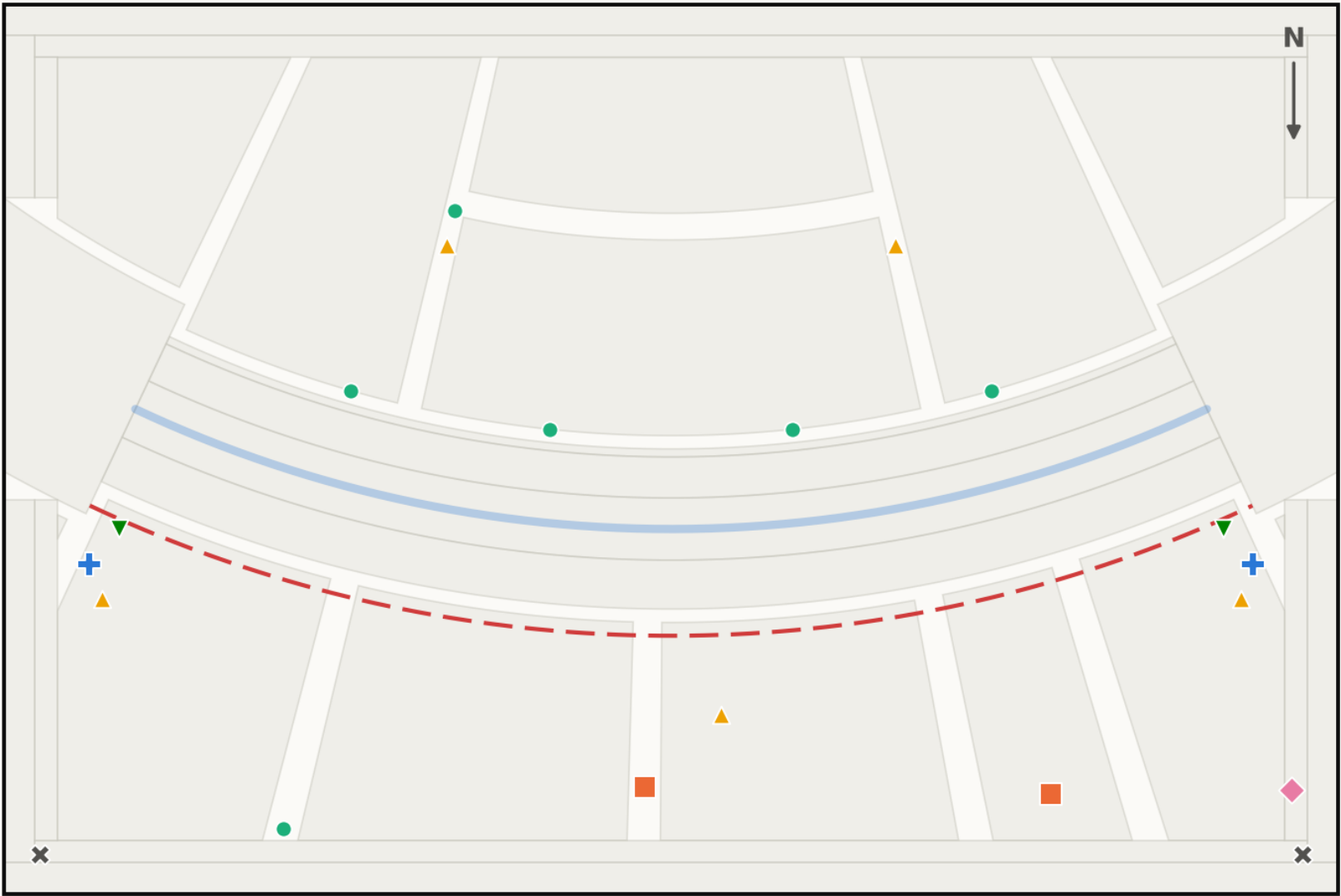
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Facilities

Technical drawing — to scale.

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



20 m

- Service / emergency vehicle access

Commercial — souk kiosks and café (2)

Public restrooms (2)
- Drinking fountain (6)

Waste & recycling station (5)

Service & maintenance store (1)
- Bicycle parking (2)

Drop-off / pick-up bay (2)

20 facilities. Commercial floor sits on the convex south face (417 m2, 2.8% of the site), facing the plaza and the event lawn so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk.

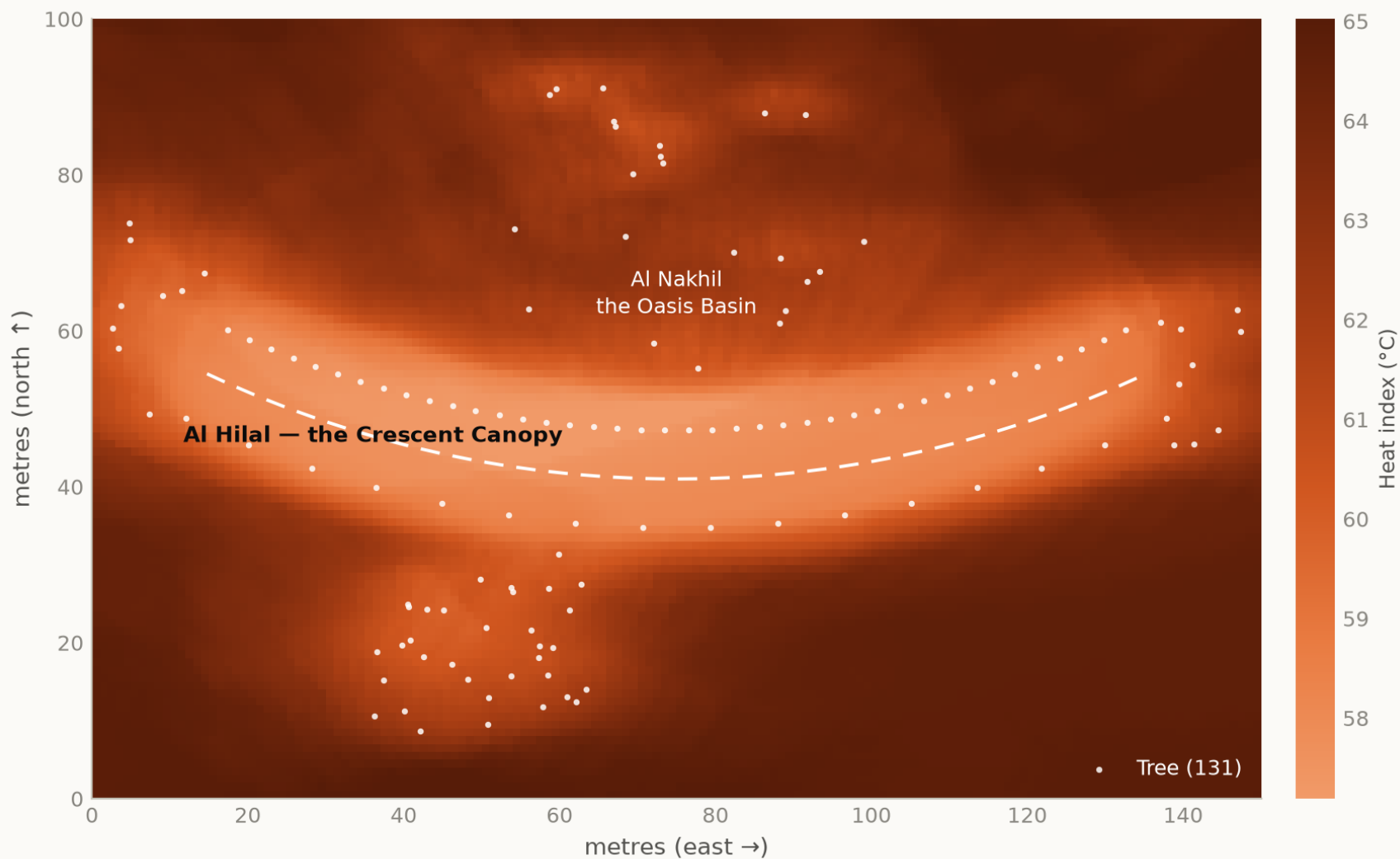
Source: src/plan.py — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig04 site comfort map

Analysis output — computed from project data.

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



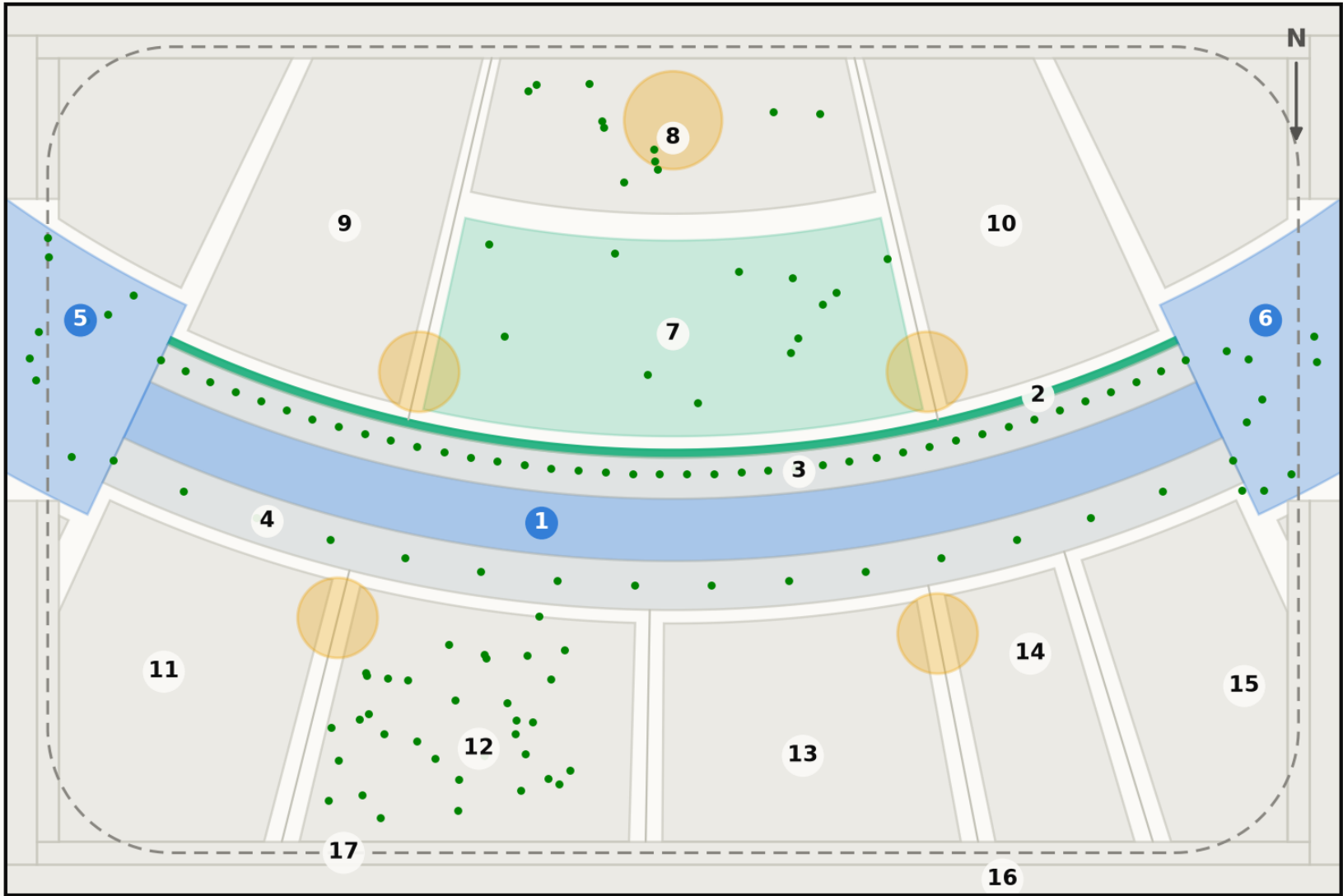
Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.
Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
Al Sikkak — shaded alleys & setbacks		1,414 m²
Total		15,000 m²

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every one of the 129 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (8)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
RENDER_PROMPTS.md	Project document
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

tools/__init__.py	Build tool
tools/build_docs.py	Build tool
tools/build_links.py	Build tool
tools/build_notebook.py	Build tool
tools/build_reports.py	Build tool
tools/build_submission_pdfs.py	Build tool
tools/organize_repo.py	Build tool
tools/report_content.py	Build tool
tools/sync_film.py	Build tool
tools/sync_portal.py	Build tool
tools/sync_submission.py	Build tool

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset
data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (9)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
archive/phases/02_PHASE2_PROBLEM_DEFINITION/README.md	Working record
archive/phases/03_PHASE3_OPPORTUNITY_AND_OBJECTIVES/README.md	Working record
archive/phases/04_PHASE4_CONCEPT_DEVELOPMENT/README.md	Working record
archive/phases/05_PHASE5_MASTERPLAN_DEVELOPMENT/README.md	Working record
archive/phases/06_PHASE6_DETAILED_DESIGN/README.md	Working record
archive/phases/07_PHASE7_PERFORMANCE_AND_SUSTAINABILITY/README.md	Working record
archive/phases/08_PHASE8_USER_EXPERIENCE_AND_ACTIVATION/README.md	Working record
archive/phases/09_PHASE9_AI_WORKFLOW_AND_VISUALIZATION/README.md	Working record

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.